



BIT CORPORATION

BIT90 SYSTEM BLOCK DIAGRAM

SHEET OF

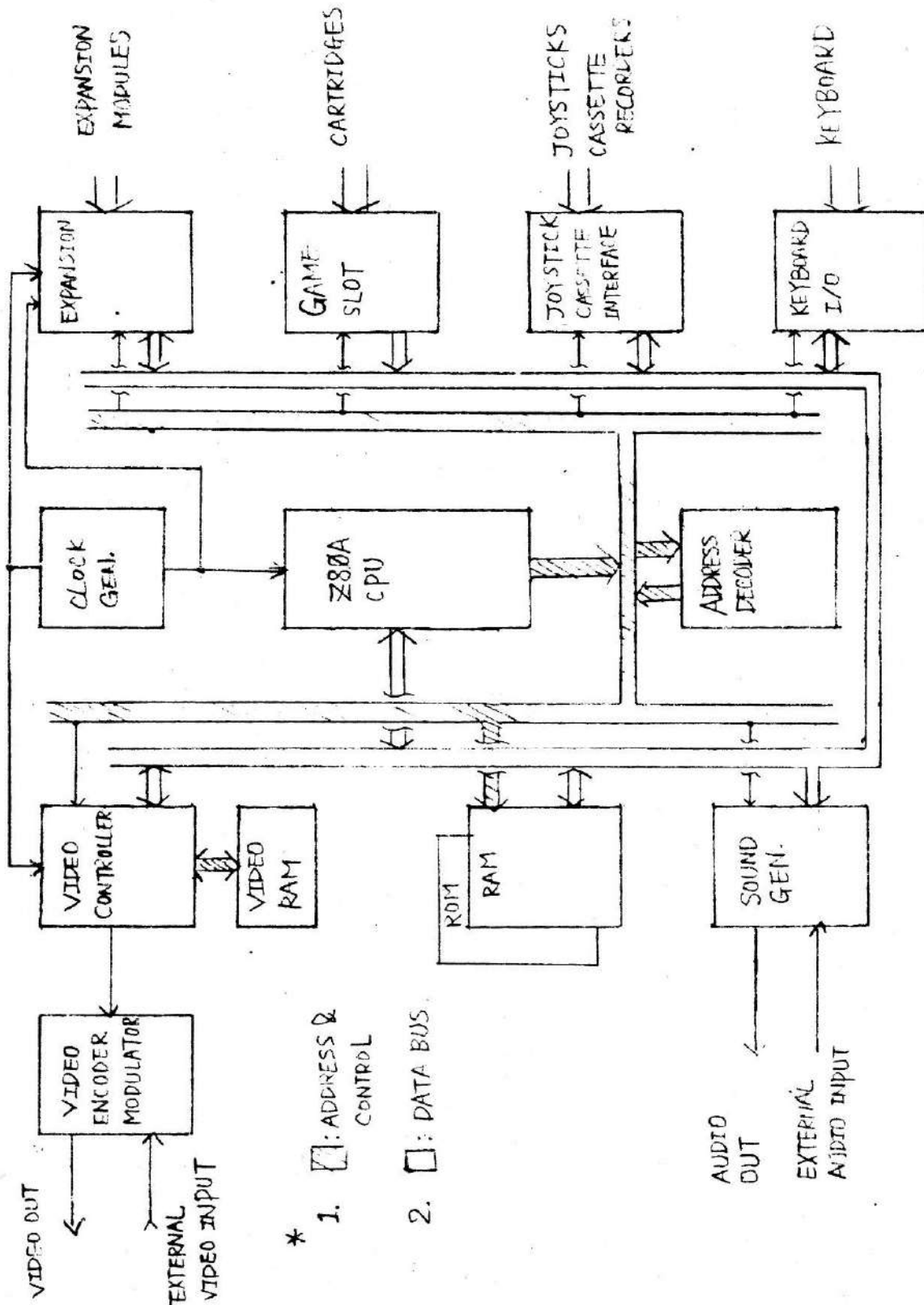
DRAWN

DATE

CHECKED

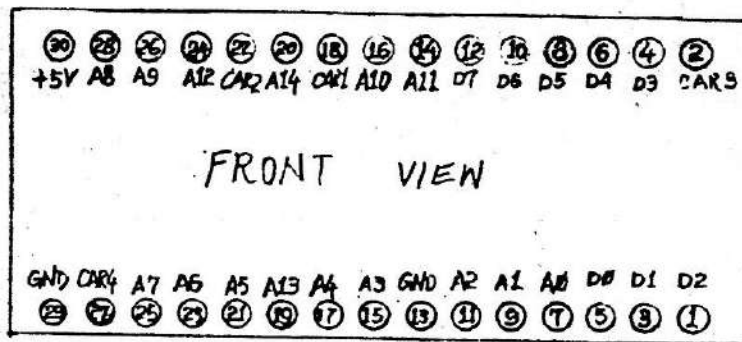
DATE

SCALE

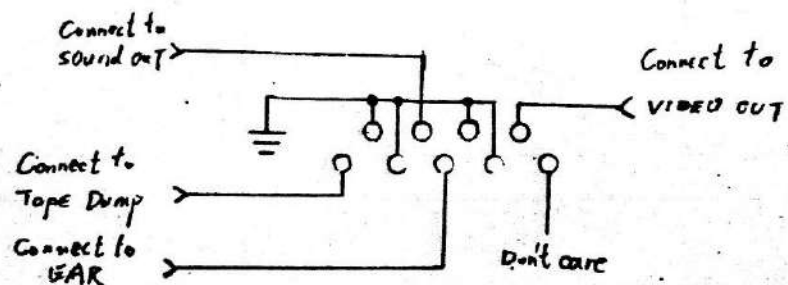


BIT90 SYSTEM BLOCK

CARTRIDGE CONNECTOR



FOR MONITOR AND RECORDER



(USE 9 PIN JACK)



BIT CORPORATION

PROCESSOR

DRAWN

DATE _____

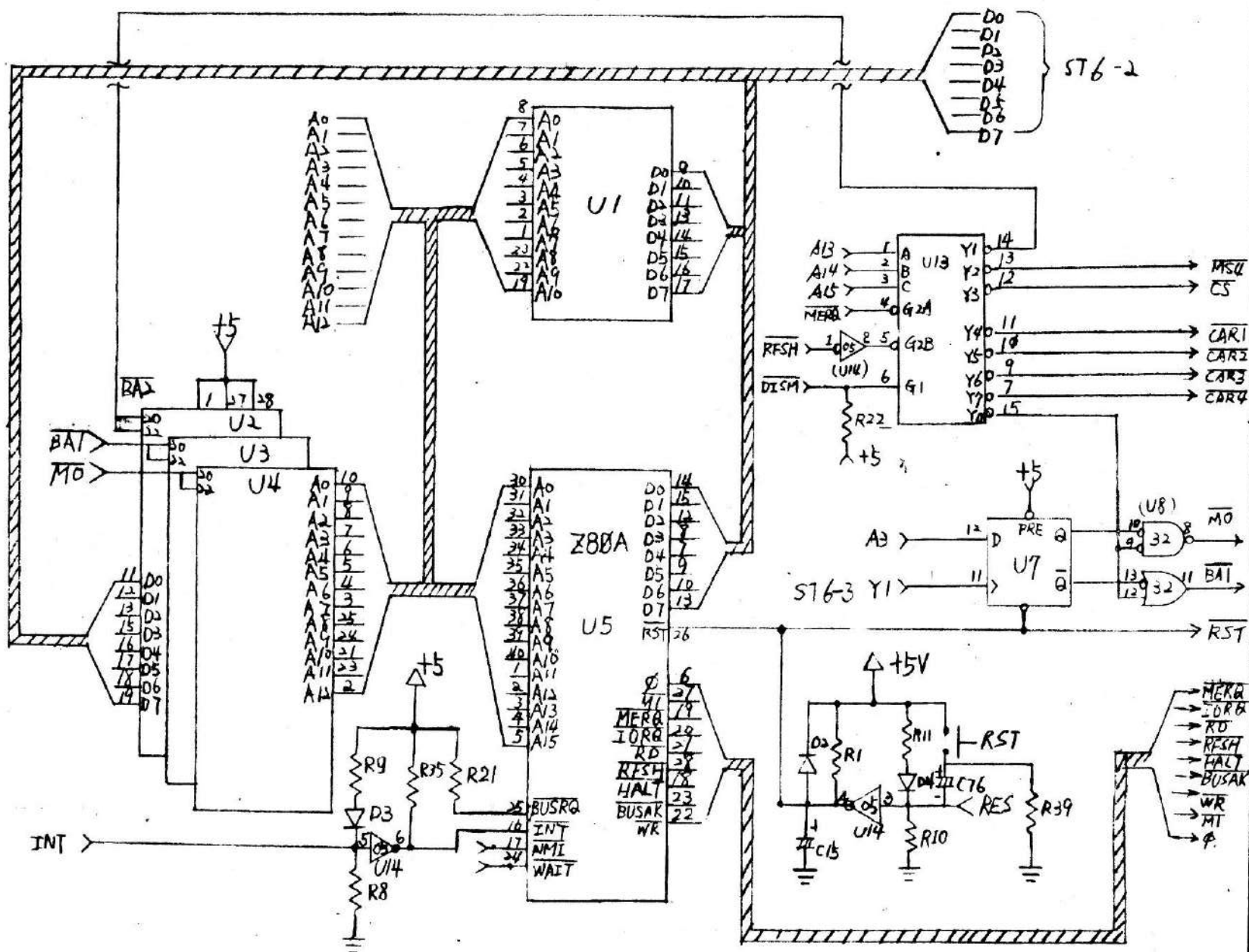
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DATE _____

SCALE

SHEET OF

6-1



SHEET 6-1 PROCESSOR



BIT CORPORATION

VIDEO CONTROL

DRAWN

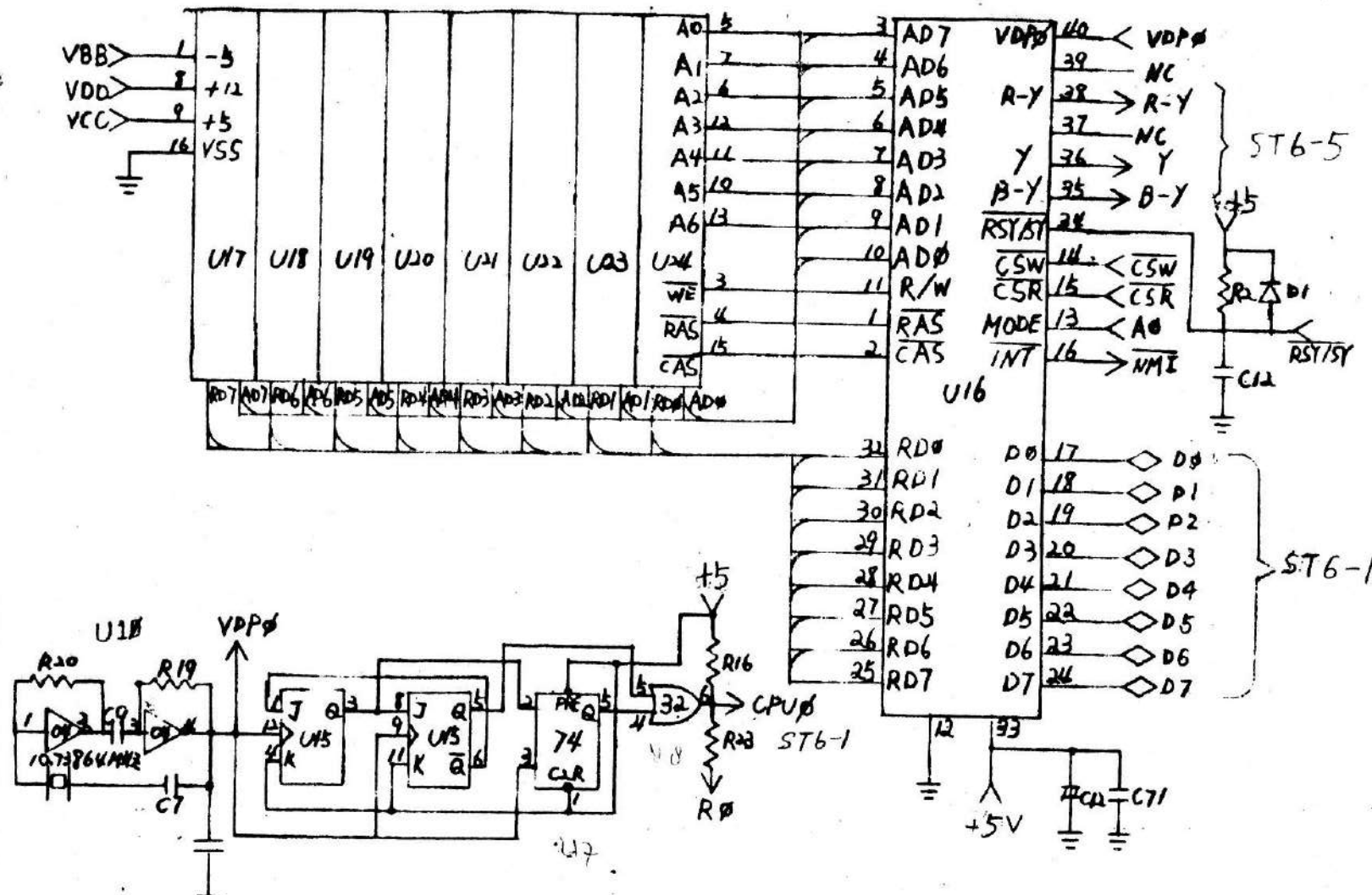
DATE

CHECKED

DATE

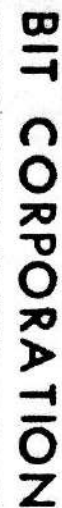
SCALE

SHEET OF 6-2



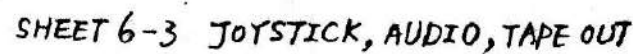
SHEET 6-2 VIDEO CONTROL

U16- TMS 9929A NL
DD4 8323
3472 PHILIPPINES



SHEET OF
6-3

SCALE





BIT CORPORATION

KEYBOARD

DRAWN

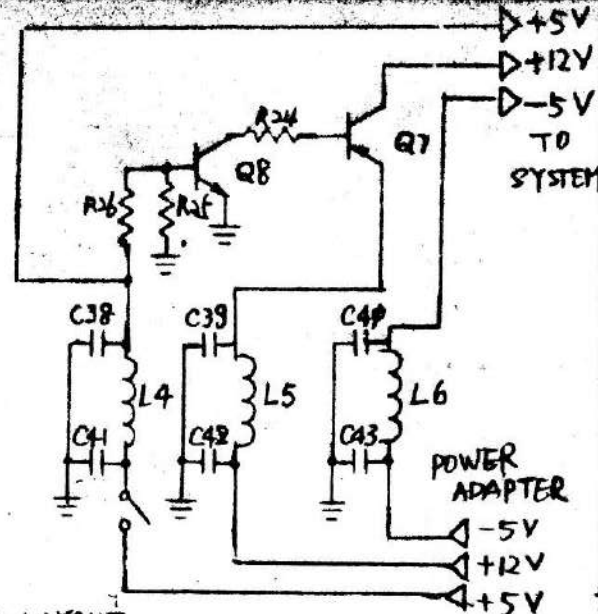
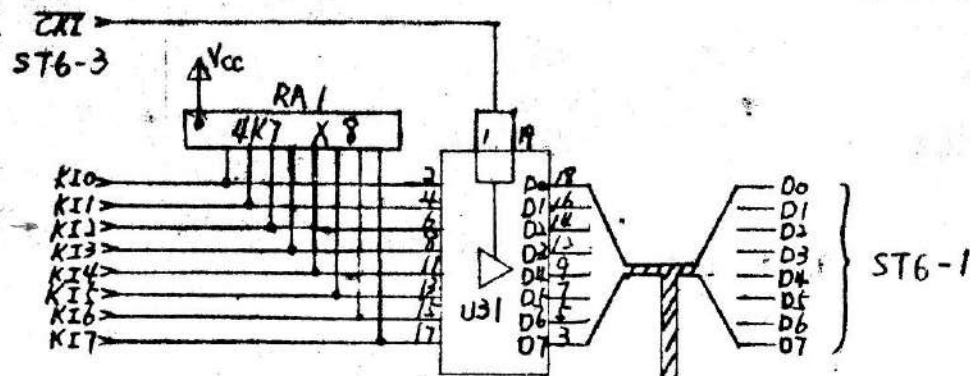
DATE

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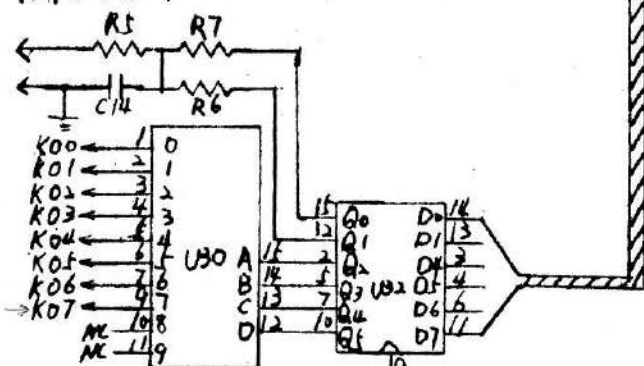
DATE

SCALE

SHEET OF 6-4



TAPE DUMP



KBC
ST6-3

OUT F8, 16 KO
IN CF, 255-2 KI

KEYBOARD LAYOUT

	1	2	4	8	16	32	64	128
0-15	=	7	3	\	INS	1	5	9
16-31	~	8	4	BS	DEL	2	6	0
32-47	P	Y	W	}	↑	ESC	R	I
48-63	{	U	E	RIGHT	→	Q	T	O
64-79	+	H	S	@	? /	CAPS LOWER	F	K
80-95	*	J	D	CR	>	A	G	L
96-111	N	V	X	M	<	Z	C	B
112-127	SHIFT	BIT	BASIC	←	↓	CTRL	SP	FC
128-143	0	1	2	3	4	5	6	7

SHEET 6-4 KEYBOARD



BIT CORPORATION

VIDEO ENCODER

DRAWN

DATE

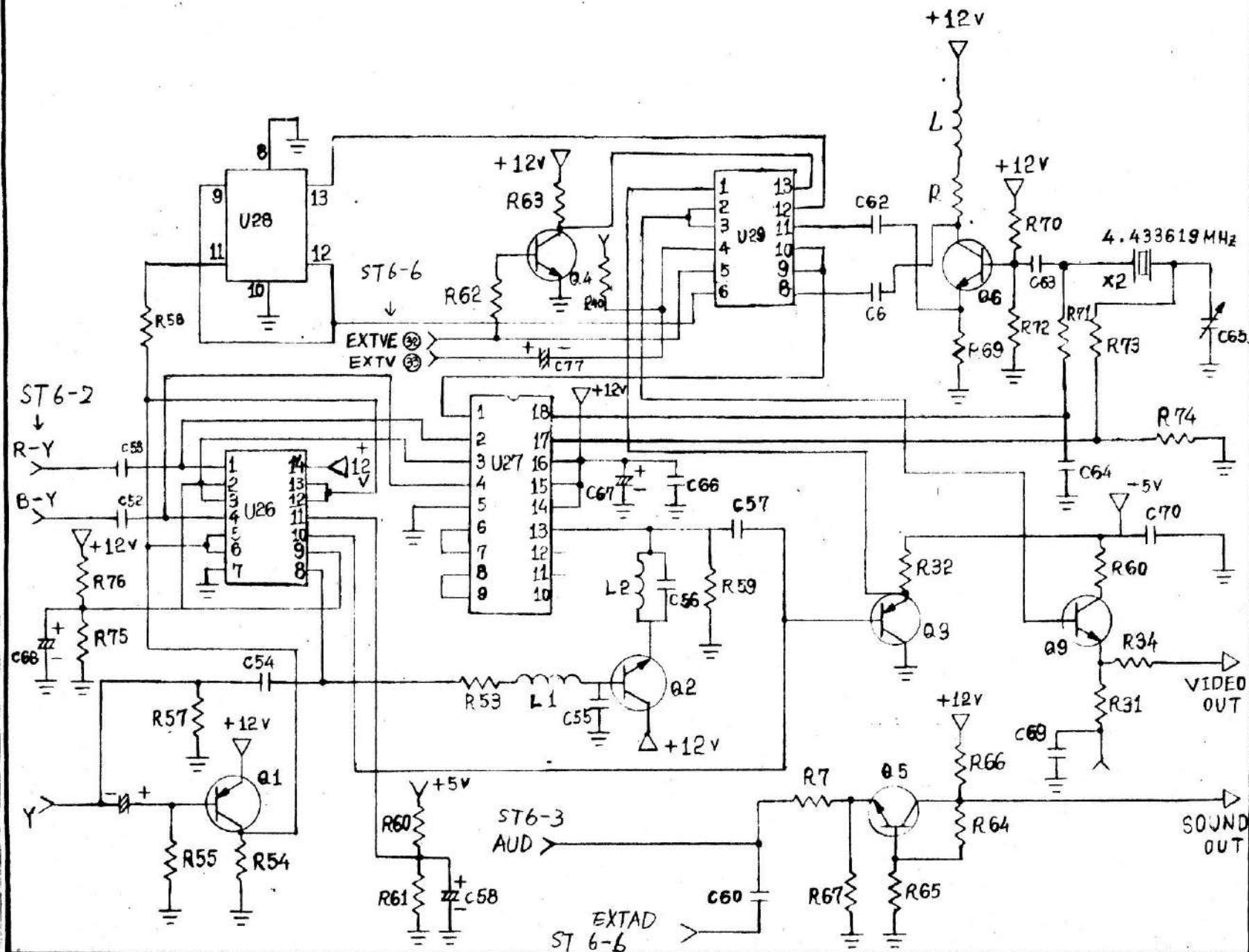
CHECKED

DATE

SCALE

SHEET OF 6-5

SHEET 6-5 VIDEO ENCODER (PAL SYSTEM)





BIT CORPORATION

EXPANSION EDGE DEFINITION

SHEET OF 6-6

DRAWN

DATE

CHECKED

DATE

SCALE

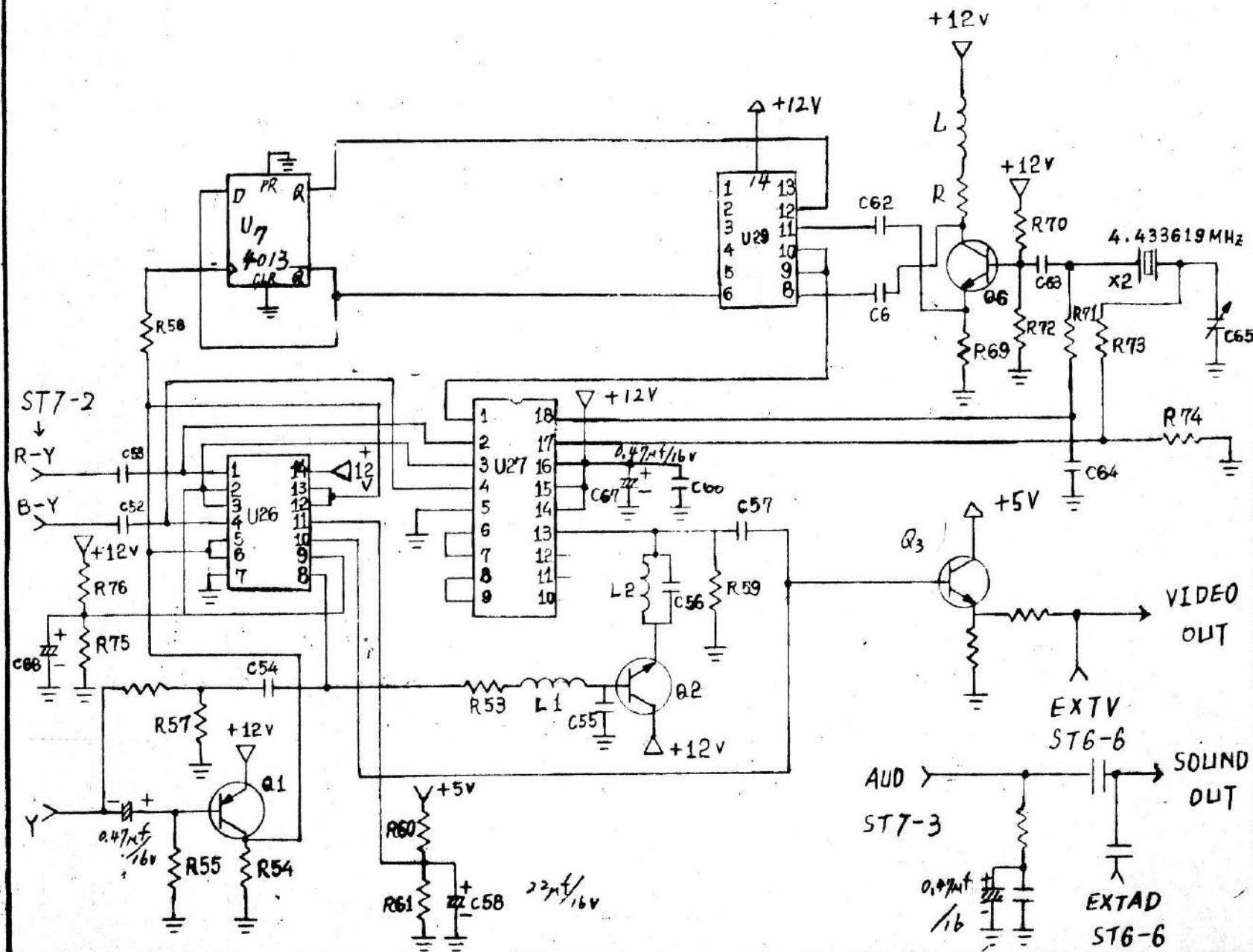
30	DISIO	31	EXTAD
29	DISM	32	EXTVE
28	A10	33	EXTV
27	A9	34	RST
26	A8	35	EXT Ø
25	A7	36	RES
24	A6	37	A11
23	A5	38	A12
22	A13	39	RST/SY
21	A4	40	R Ø
20	A2	41	NC
19	D4	42	NC
18	A1	43	A15
17	D6	44	A3
16	D7	45	CPU Ø
15	M1	46	D2
14	D0	47	A0
13	RST	48	D5
12	D1	49	RFSH
11	BUSRQ	50	WAIT
10	INT	51	INT
9	NMI	52	BUSAK
8	WR	53	RD
7	HALT	54	MERQ
6	NC	55	IORQ
5	MS4	56	AUDIO OUT
4	A14	57	+12 V
3	D3	58	+5 V
2	GND	59	+5 V
1	GND	60	-5 V

ST6-6 EXPANSION EDGE
DEFINITION

SHEET 7-5 VIDEO ENCODER (PAL SYSTEM)



BIT CORPORATION



RAW

DATE

CHECKED

DATE

SCALE

VIDEO ENCODER

/32k

PAL

SHEET OF 7-5



BIT CORPORATION

32K DRAM (4416 X 4)

SHEET OF

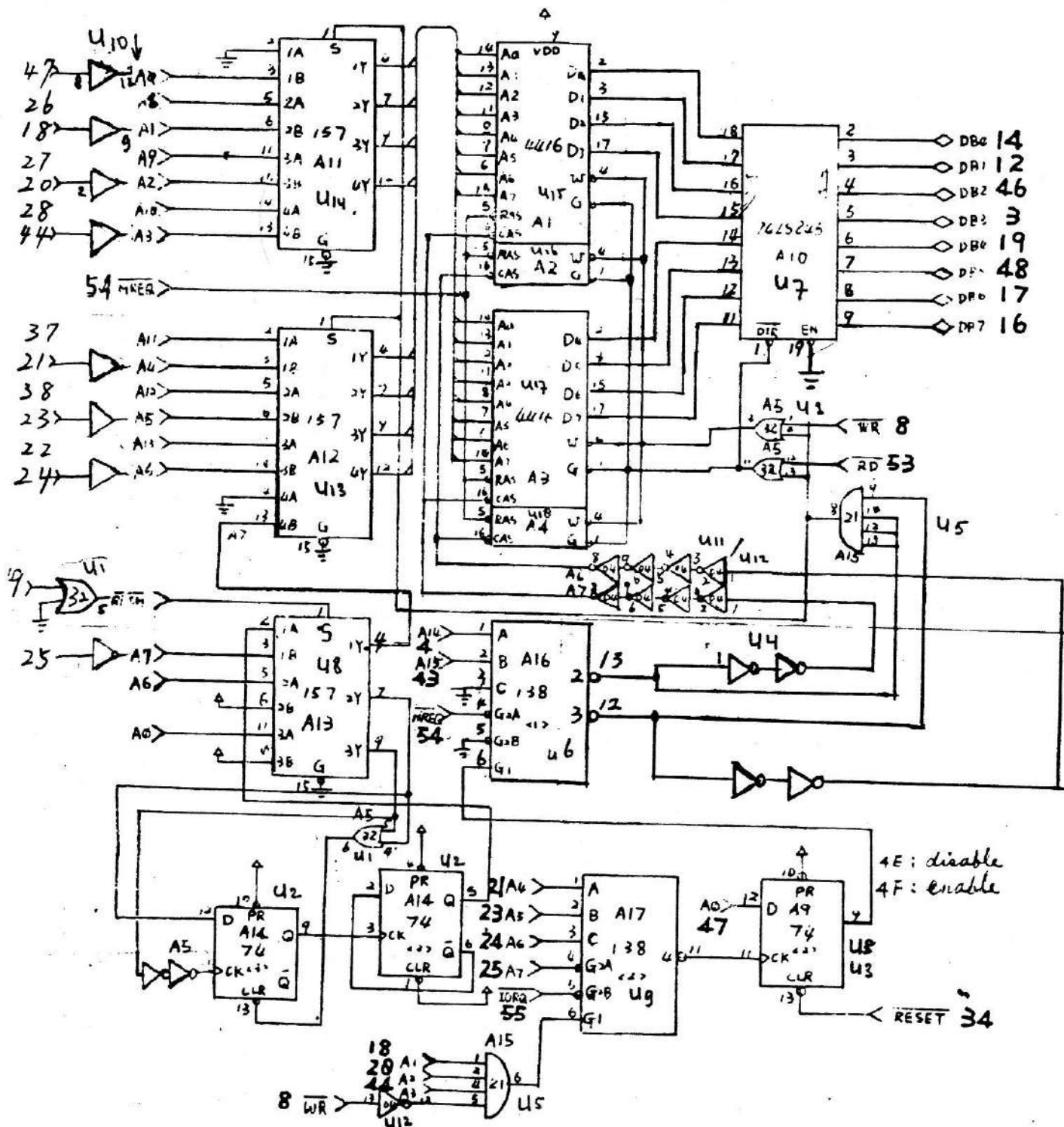
RAWN

DATE

CHECKED

DATE

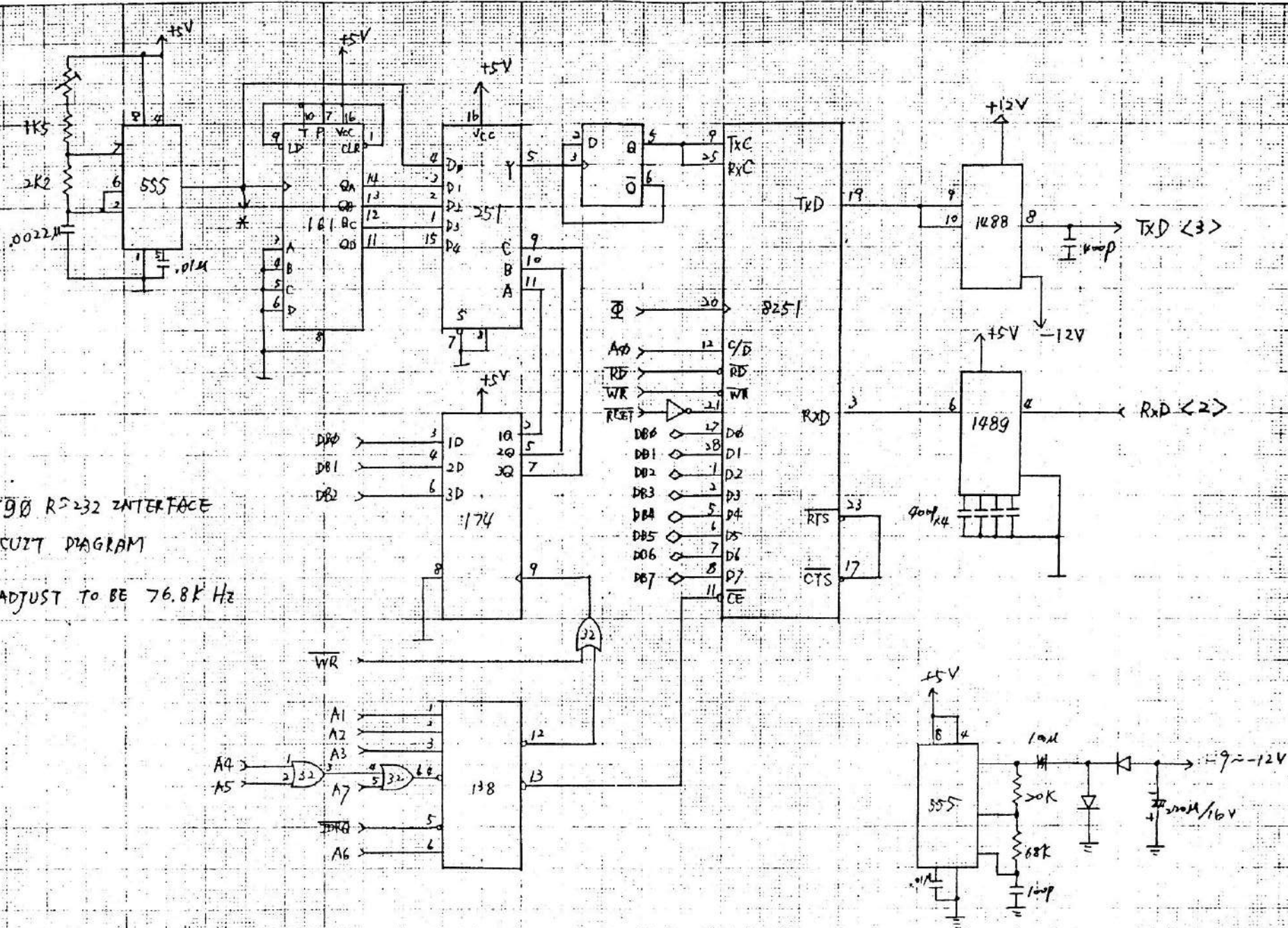
SCALE



普泽有限公司

BIT90 RS232 INTERFACE CIRCUIT DIAGRAM

* ADJUST TO BE 76.8K Hz





BIT CORPORATION

BIT90 PARTS LIST (PAL SYSTEM)

SHEET OF

DRAWN

DATE

CHECKED

DATE

SCALE

R	
1	10 K
2	1 K
3	270 Ω
4	470 Ω
5	10 K
6	22 K
7	56 K
8	1 K
9	100 K
10	1 K
11	100 K
12	15 K
13	1 K
14	10 K
15	1.2 K
16	330 Ω
17	220 Ω
18	4.7 K
19	470 Ω
20	470 Ω
21	3.3 K
22	3.3 K
23	470 Ω
24	1 K
25	100 K
26	10 K
30	220 Ω
40	27 K
32	47 Ω
33	4.7 K
34	10 Ω
35	3.3 K
36	3.3 K
37	3.3 K
38	3.3 K
39	3.3 K
51	390 Ω
52	390 Ω
53	3.3 K
54	2.7 K
55	120 K
57	100 Ω
58	27 K
59	1.8 K
60	1.2 K
61	430 Ω

62	10 K
63	10 K
64	22 K
65	10 K
66	6.8 K
67	330 Ω
68	390 Ω
69	330 Ω
70	2.7 K
71	1.5 K
72	1 K
73	4.7 K
74	2.7 K
75	1 K
76	1 K

D	
1	←
2	←
3	1N4148
4	←
5	←
6	←

Q	
1	2N2907
2	2N2222
3	2N2907
4	2N2222
5	2N2222
6	2N2222
7	A684
8	2N2222
9	2N2222
X1	10.738635 MHz
X2	4.433619 MHz

R	
RA1	4.7 K
2	10 K
3	10 K

C	
1	←
2	←
3	0.1
4	←
5	←
6	←
7	56 P
8	0.1
9	103
10	0.1
11	104
12	10 μ F/16V
13	0.1
14	0.002 K
15	0.47 μ F/50V
16	←
18	←
19	←
20	←
21	←
22	←
23	←
24	←
25	←
26	←
27	0.1
28	←
29	←
30	←
31	←
32	←
33	←
34	←
35	←
36	←
37	←
38	←
39	←
40	←
41	←
42	←
51	0.47 μ F/50V

P A L

52	0.1
53	0.1
54	0.1
55	33 P
56	47 P
57	0.1
58	10 μ F/16V
60	107
61	0.1
62	0.1
63	33 P
64	33 P
65	5~20 P
66	473 P
67	0.47 μ F/50V
68	10 μ F/16V
69	←
70	←
71	←
72	0.1
73	←
74	←
75	←
76	0.47 μ F/50V
77	10 μ F/16V
78	33 P

L

1	39 μ H
2	15 μ H
3	4.7 μ H
4	12 μ H
5	12 μ H
6	6 μ H

U

1	TC5517
2	2764
3	2764
4	2764
5	280A CPU
6	76489
7	74LS74
8	74LS32
9	74LS00
10	74LS04
11	74LS138
12	74LS00
13	74LS138
14	74LS05
15	74LS107
16	(VDP)
17	←
18	←
19	←
20	4116
21	←
22	←
23	←
24	←
25	LM393
26	CD4066
27	LMJ889
28	CD4013
29	CD4066
30	74LS145
31	74LS244
32	74LS174
33	74LS244
34	74LS244

CA1	18 PIN MOLEX (MALE)
CN2	30 PIN CONNECTOR (FEMALE)
TM1	9 PIN JACK
TM2	9 PIN JACK
TM3	9 PIN JACK
PJ1	5 PIN POWER JACK
PS	W1 power switch
RF	RCA
J1	JACK
	HEAT SINK (FOR VDP)

RS232C INTERFACE CARD

BIT90 computer has an expansion module for communication with any other computers or terminals. This module can be set four speeds and two operation modes. When you connect BIT90 computer with RS232C interface card; first, you should select the transmission or receive mode, next, set the baud rate for communication. It is very important to set the same speed between your BIT90 computer and the other side device. Of course, the program built in cassette must be loaded into computer and executed by BIT90. The following steps tell you how to set up this module.

- Step 1: Load the cassette tape from recorder to computer. You may type a command "LOAD:RUN", then computer will execute the program automatically.
- Step 2: When the program has been executed, the screen will display the message as shown below:
- 1. TRANSMISSION
 - 2. RECEIVE
 - (PRESS FCNT 1 TO RESTART)
 - SELECT MODE:
- Step 3: If you have selected the communication mode then the screen will display the following message:
- 1. 2400 BPS
 - 2. 1200 BPS
 - 3. 600 BPS
 - 4. 300 BPS
 - SELECT SPEED?
- Step 4. In step 3, you should set the BAUD RATE for communication speed. Generally 300 BPS low speed is used to communicate with TTY or telephone modem. Once you have set the speed, you may begin to communicate with other terminals or computers. Be sure that the speed you have set must be the same as the others when you are talking to each other.

SPECIFICATION

After connecting BIT 90 home computer with RS232C Interface, please key in a program as next page indicated.

PROCESS

STEP 1. Select mode (1. Transmission 2. Receive)

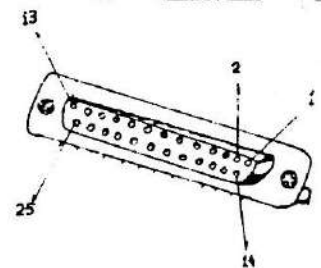
STEP 2. Select speed (Notice: It is very important to set the same speed between your BIT 90 computer and the other side device.)

In this program, Line 30 initialize the Baud Rate and RS232C Interface module; "WAIT" statement in Line 45 is used to await 'Transmission' and in Line 56 is used to await 'Receive'.

DEFINITION

<u>PIN</u>	<u>DESCRIPTION</u>
2	RxD(Receive Data)
3	TxD(Transmission Data)
7	Signal Ground

ILLUSTRATION



<u>I/O Port</u>	<u>DESCRIPTION</u>
68	Data I/O
69	Command or Read status
70	Select Baud Rate

25 PIN D TYPE CONNECTOR

B2790 TEST PROGRAM FOR RS332 INTERFACE MODULE

READY

>LIST

```

5  HOME :PRINT :PRINT " RS232C INTERFACE"
10 PRINT :PRINT :PRINT " 1. TRANSMISSION":PRINT " 2. RECEIVE":PRINT "
(PRESS FCTN 1 TO RESTART)"
15 PRINT :INPUT " SELECT MODE: ",K:PRINT :PRINT
20 PRINT " 1. 2400 BPS":PRINT " 2. 1200 BPS":PRINT " 3. 600 BPS":PRINT " 4
300 BPS":PRINT
25 INPUT " SELECT SPEED?",S
30 OUT 70,S-1:OUT 69,0:OUT 69,0:OUT 69,0:OUT 69,64:OUT 69,206:OUT 69,55
35 HOME :IF K=2GOTO 50
40 PRINT " MESSAGE FOR TRANSMISSION";CHR$( 13)
42 IF ASC( INKEY$ )=255GOTO 42ELSE IF ASC( INKEY$ )=225GOTO 5
43 A#=INKEY$ :PRINT A#;:IF ASC( A#)=13PRINT
45 WAIT 69,1,0:OUT 68,ASC( A#):CLEAR
46 IF ASC( INKEY$ )<> 255GOTO 46ELSE GOTO 42
50 PRINT " RECEIVED DATA:"
55 IF ASC( INKEY$ )=225GOTO 5
56 WAIT 69,2,0:A=IN( 68):IF A>128THEN A=A-128
57 A#=CHR$( A):PRINT A#;:OUT 68,ASC( A#):IF ASC( A#)=13PRINT
58 CLEAR :GOTO 55

```

READY

>

READY

>



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台北郵政信箱74之17號
電話: (02) 731-2838~9 • 731-1368~9
954-0854

Control Byte for BIT90 RS232 Interface

DATE:

PG

1. PORT 70 IS USED FOR SPEED SELECTION:

S-1	SPEED		
	MODE = 206	MODE = 207	MODE = 205
0	2400 Band	600 Band	38400 Band
1	1200 "	300 "	19200 "
2	600 "	150 "	9600 "
3	300 "	75 "	4800 "
4	150 "		2400 "

2. LINE 30 ILLUSTRATION:

OUT 70, S-1 : REM SPEED SELECTION

OUT 69, 0 : REM 3 NULL 'OUT' FOR INITIALIZATION

OUT 69, 0

OUT 69, 0

OUT 69, 64 : REM SOFTWARE RESET USART

OUT 69, 206 : REM MODE INSTRUCTION

OUT 69, 55 : REM COMMAND INSTRUCTION

USART-8251 PROGRAMMING MODEL

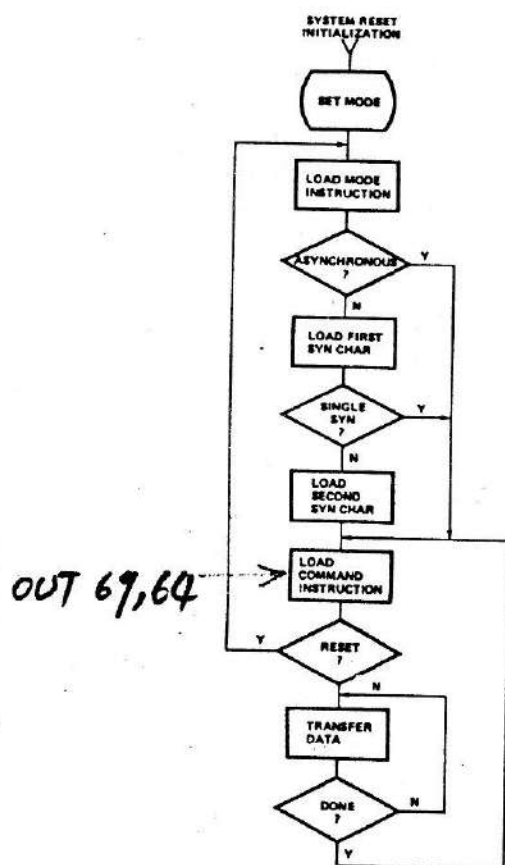


Figure 4. Initialization Flowchart

register. The mode control register is loaded by the first control output ($C/\bar{D}=1$, $RD=1$, $WR=0$, $\overline{CS}=0$) following a reset. The format of the mode control instruction is shown in Figure 5. The instruction can be considered as four 2-bit fields. The first 2-bit field ($D_1 D_0$) determines whether the USART is to operate in the synchronous (00) or asynchronous mode. In the asynchronous mode this field also controls the clock scaling factor. As an example, if D_1 and D_0 are both ones, the RxC and TxC will be divided by 64 to establish the baud rate. The second field, D_3-D_2 , determines the number of data bits in the character and the third, D_5-D_4 , controls parity generation. Note that the parity bit (if enabled) is added to the data bits and is not considered as part of them when setting up the character length. As an example, standard ASCII transmission, which is seven data bits plus even parity, would be specified as:

X X 1 1 1 0 X X

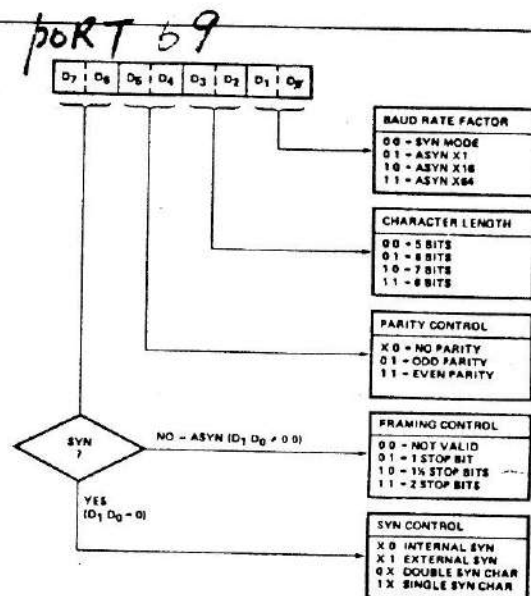


Figure 5. Mode Instruction Format

The last field, D_7-D_6 , has two meanings, depending on whether operation is to be in the synchronous or asynchronous mode. For the asynchronous mode (i.e., $D_1 D_0 \neq 00$), it controls the number of STOP bits to be transmitted with the character. Since the receiver will always operate with only one STOP bit, D_7 and D_6 only control the transmitter. In the synchronous mode ($D_1 D_0 = 00$), this field controls the synchronizing process. Note that the choice of single or double SYN characters is independent of the choice of internal or external synchronization. This is because even though the receiver may operate with external synchronization logic, the transmitter must still know whether to send one or two SYN characters should the CPU fail to supply a character in time.

Following the loading of the mode instruction the appropriate SYN character (or characters) must be loaded if synchronous mode has been specified. The SYN character(s) are loaded by the same control output instruction used to load the mode instruction. The USART determines from the mode instruction whether no, one, or two SYN characters are required and uses the control output to load SYN characters until the required number are loaded.

At completion of the load of SYN characters (or after the mode instruction in the asynchronous mode), a command character is issued to the USART. The command instruction controls the operation of the USART within the basic framework established by the mode instruction. The format of the command instruction is shown in

USART-8251 PROGRAMMING MODEL

Figure 6. Note that if, as an example, the USART is waiting for a SYN character load and instead is issued an internal reset command, it will accept the command as a SYN character instead of resetting. This situation, which should only occur if two independent programs control the USART, can be avoided by outputting three all zero characters as commands before issuing the internal reset command. The USART indicates its state in a status register which can be read under program control. The format of the status register read is shown in Figure 7.

When operating the receiver it is important to realize that RxE (bit 2 of the command instruction) only inhibits the assertion of RxRDY; it does not inhibit the actual reception of characters. Because the receiver is constantly running, it is possible for it to contain extraneous data when it is enabled. To avoid problems this data should be read from the USART and discarded. The read should be done immediately following the setting of Receive Enable in the asynchronous mode, and following the setting of Enter Hunt in the synchronous mode. It is not necessary to wait for RxRDY before executing the dummy read.

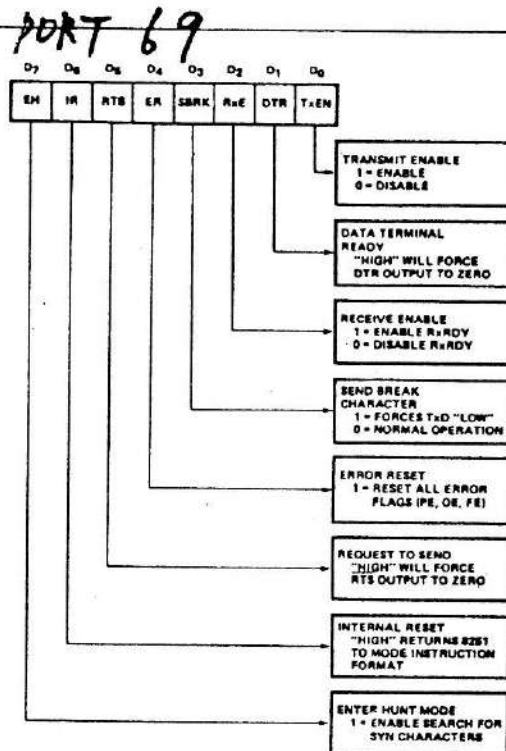
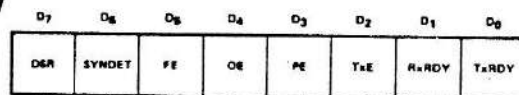


Figure 6. Command Instruction Format

PORT 69



SAME DEFINITIONS AS I/O PINS EXCEPT THAT TxRDY IS NOT CONDITIONED BY TxE OR CTE.

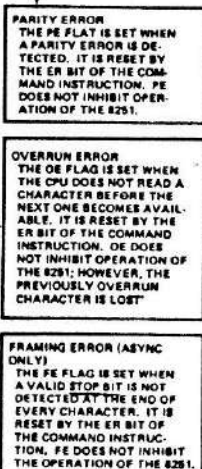


Figure 7. Status Register Format

• FOR TRANSMISSION

WAIT 69, 1, 0

• FOR RECEIVE

WAIT 69, 2, 0

INSTRUCTION MANUAL FOR

BIT90

DOT MATRIX PRINTER

 **BIT CORPORATION**

Instruction Manual for 40-column Printer

For the purpose of recording the screen of BIT90 computer there is a printer designed specially for BIT90. You are requested to read this booklet very carefully before using this printer.

1. SPECIFICATIONS

Printing method: Serial Impact Dot MATRIX.
Printing format: 5×7 Dot MATRIX.
Printing speed: 40 CPS
Printing direction: Single direction. left to right.
Columns/line: 40 columns
Paper feed: Friction Feed only
Paper type: Common paper. 69mm ± 1mm.
Ribbon: Standard typewriter ribbon black & red
Print head life: Approximately 40 million characters (replaceable)

2. PREPARATION FOR USE.

2-1 Unpacking, counting the parts.

Before you unpack your printer inspect the carton for sign of damage. If it appears to be damaged, be especially careful when you inspect its contents. The package should contain following.

1. This user's manual.
2. Printer unit with ribbon, paper and flat cable.
3. Interface module.

If any of this three is missing or damaged, notify your dealer immediately.

2-2. Setting up

Before starting to use your printer, you should make sure that it is in good, connect with your computer. So follow this procedure

1. Lift and remove the plastic top cover to expose print head and mechanism.
2. Remove shipping tape.
3. Plug the interface module into the expansion slot of BIT90, and make sure the interface module's edge connector is fixed with slot's pin. Before you do this, you must turn BIT90 power off otherwise computer or interface module will get damaged.
4. Turn on the power of computer & printer, see if the printer's power LED is on or not. If not, check your power system, and if it still not work, notify your dealer immediately.

3. FUNCTION DESCRIPTION

The BIT90 printer module supports 4 modes of printing, namely TEXT, HI-RESOLUTION, LO-RESOLUTION, BIT-IMAGE. The TEXT, HI-RES, LO-RES are all Hard Copy mode; but the BIT-IMAGE

—2—

mode will not show pattern on screen. To switch between these 4 modes, simply send their control code (control code is not echoed on screen), and the control code can be redefined by POKEing in BASIC.

4. OPERATION

4-1. TEXT MODE.

Turn off the power of BIT90, then connect printer correctly, turn on power, after a moment when computer is in BASIC, type "CALL 30720", from now on, the printer is ready to print. After the procedure described above, printer is now in TEXT mode.

(EXAMPLE) CALL 30720 (CR)

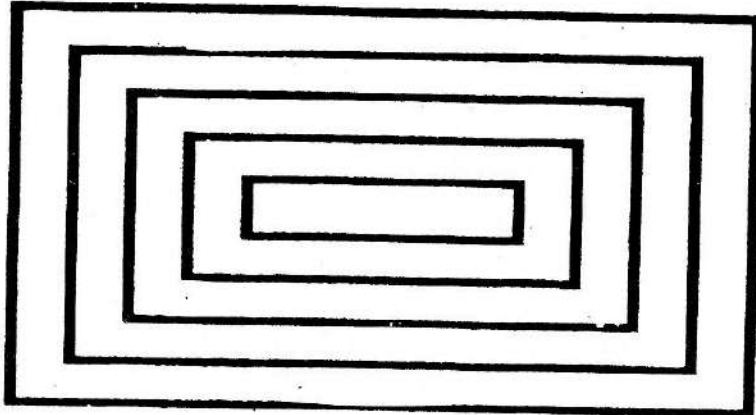
BIT90 
Graphic & Text PRINTER

READY

4.2. LO-RES MODE

While in TEXT mode, sent a control code 05H (by means of "PRINT CHR\$(5)" as in BASIC), as soon as printer receives the code, the full screen of low resolution (48 x 64) will be dumped.

```
10 REM LO-RESOLUTION
20 CALL SCREEN(1,2)
30 DX=0:OY=0
40 BS=5
50 FOR SP=47 TO 0 STEP -BS-BS
60 PLOT OY,DX,15 TO OY,DX+SP+16 TO OY+SP,DX+SP+16 TO OY+SP,DX TO OY,DX
70 DX=DX+BS:OY=OY+BS
80 NEXT SP
90 PRINT CHR$( 5):REM LO-RESOLUTION
00 PRINT CHR$(27):TURN OFF PRINTER
```

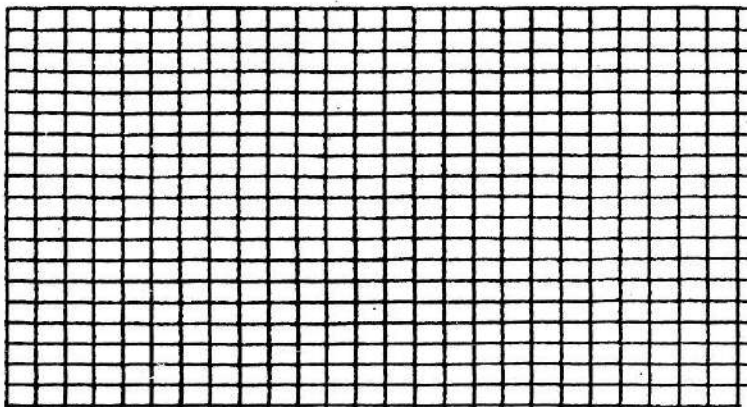


-4-

4.3. HI-RES MODE

As described in LO-RES, except the code is 02H and the high resolution screen is dumped.

```
10 REM HI-RESOLUTION
20 CALL SCREEN(1,1)
30 PLOT 0,0,15
40 FOR X=0 TO 255 STEP 10
50 PLOT 0,X TO 191,X
60 NEXT X
70 FOR Y=0 TO 191 STEP 10
80 PLOT Y,0 TO Y,255
90 NEXT Y
100 PRINT CHR$( 2):REM HI-RESOLUTION
```



-5-